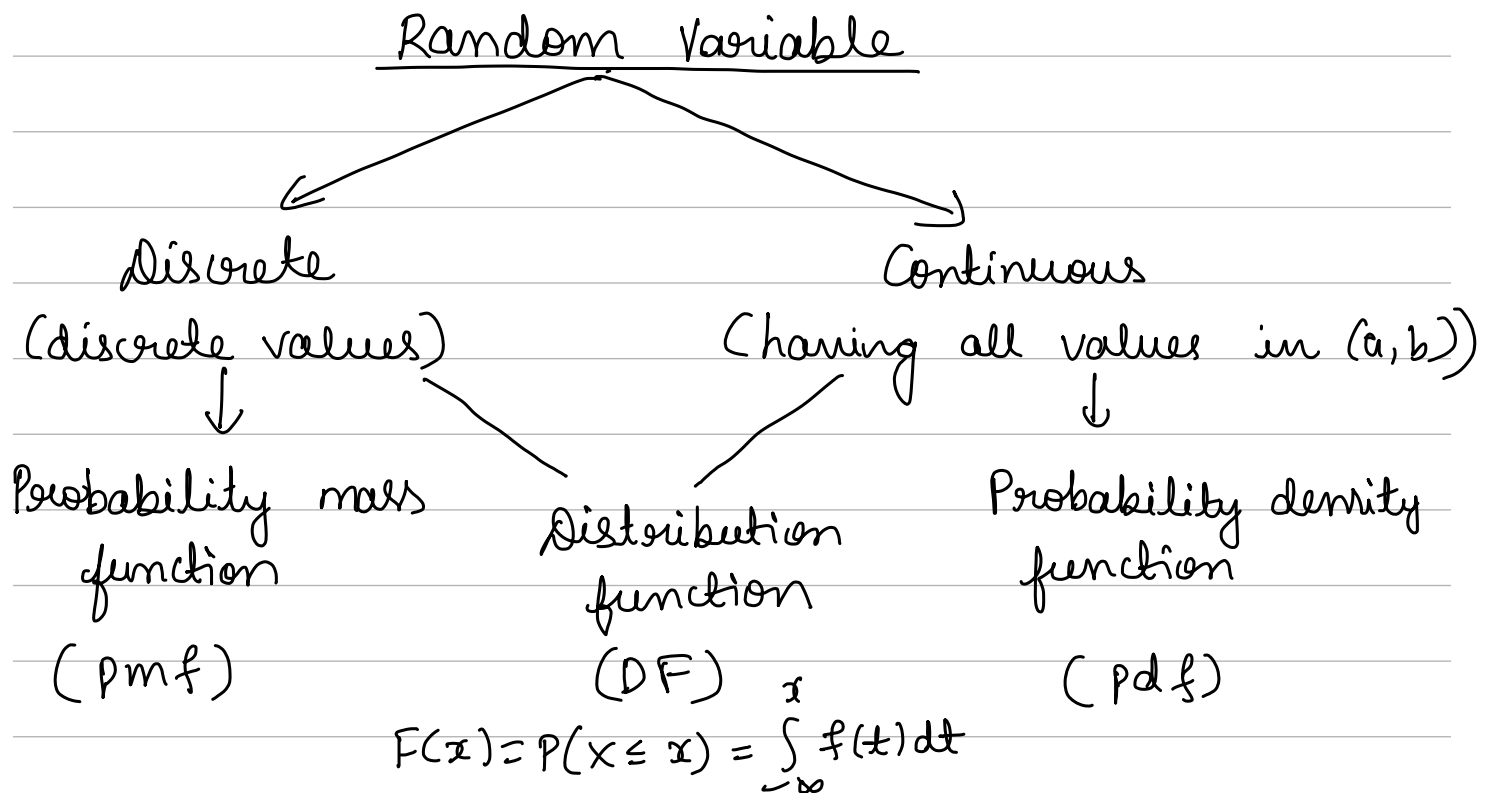


Problems on distribution function.



1) If X is a r.v with pdf

$$f(x) = \begin{cases} e^{-x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

Find distribution function

Solⁿ:- $F(x) = P(X \leq x) = \int_{-\infty}^x f(x) dx$

$$= \int_{-\infty}^0 f(x) dx + \int_0^x f(x) dx$$

$$= 0 + \int_0^x e^{-x} dx$$

$$= -e^{-x} + 1 = 1 - e^{-x}$$

$$\therefore F(x) = \begin{cases} 0, & x < 0 \\ 1 - e^{-x}, & x > 0 \end{cases}$$

$$2) \text{ If } p(x) = \begin{cases} \frac{x}{15}, & x = 1, 2, 3, 4, 5 \\ 0, & \text{otherwise} \end{cases}$$

$$a) \text{ Find } P(X=1 \text{ or } X=2)$$

$$b) P\left(\frac{1}{2} < X < \frac{5}{2} \mid X > 1\right)$$

Solⁿ

$$\begin{aligned} a) P(X=1 \text{ or } X=2) &= P(X=1) + P(X=2) \\ &= P(1) + P(2) \\ &= \frac{1}{15} + \frac{2}{15} = \frac{1}{5} \end{aligned}$$

$$\begin{aligned} b) P\left(\frac{1}{2} < X < \frac{5}{2} \mid X > 1\right) &= \frac{P\left(\left(\frac{1}{2} < X < \frac{5}{2}\right) \cap (X > 1)\right)}{P(X > 1)} \quad \left\{ \begin{array}{l} \text{using} \\ P(A|B) = \frac{P(A \cap B)}{P(B)} \end{array} \right. \end{aligned}$$

$$= \frac{P(\{1, 2\} \cap \{2, 3, 4, 5\})}{P(2, 3, 4, 5)}$$

$$= \frac{P(2)}{P(2, 3, 4, 5)}$$

$$\begin{aligned}
 &= \frac{P(2)}{P(2) + P(3) + P(4) + P(5)} \\
 &= \frac{\frac{2}{15}}{\frac{2}{15} + \frac{3}{15} + \frac{4}{15} + \frac{5}{15}} \\
 &= \frac{2}{14} = \frac{1}{7}
 \end{aligned}$$

3. If X is a conv with pdf

$$f(x) = \begin{cases} c(4x - 2x^2), & 0 < x < 2 \\ 0, & \text{otherwise} \end{cases}$$

- find c
- Find $P(X > 1)$
- Find DF

Solⁿ

a) We know $\int_{-\infty}^{\infty} f(x) dx = 1$

$$\begin{aligned}
 \int_{-\infty}^{\infty} f(x) dx &= \int_{-\infty}^0 f(x) dx + \int_0^2 f(x) dx + \int_2^{\infty} f(x) dx \\
 &= \int_0^2 f(x) dx = \int_0^2 c(4x - 2x^2) dx = 1
 \end{aligned}$$

$$\Rightarrow 4c \left. \frac{x^2}{2} \right|_0^2 - 2c \left. \frac{x^3}{3} \right|_0^2 = 1$$

$$\Rightarrow 4c \left(\frac{4}{2} \right) - 2c \left(\frac{8}{3} \right) = 1$$

$$\Rightarrow 8c - \frac{16c}{3} = 1$$

$$\Rightarrow 24c - 16c = 3$$

$$\Rightarrow 8c = 3 \Rightarrow c = \frac{3}{8}$$

$$b) P(X > 1) = 1 - P(X \leq 1)$$

$$= 1 - \int_0^1 f(x) dx$$

$$= 1 - \int_0^1 \left(\frac{3}{8} \right) (4x - 2x^2) dx$$

$$= 1 - \int_0^1 \frac{3}{4} (2x - x^2) dx$$

$$= 1 - \frac{3}{4} \left. \frac{2x^2}{2} \right|_0^1 + \frac{3}{4} \left. \frac{x^3}{3} \right|_0^1$$

$$= 1 - \frac{3}{4} (1) + \frac{1}{4} (1)$$

$$= 1 + \left(\frac{1-3}{4} \right) = 1 - \frac{1}{2} = \frac{1}{2}$$

c) To find D.F :-

$$-\infty < x \leq 0, \quad f(x) = 0 \Rightarrow F(x) = 0$$

$$\begin{aligned}
 0 < x < 2, \quad F(x) &= \int_0^x \frac{3}{8} (4x - 2x^2) dx \\
 &= \frac{3}{8} \left(4 \frac{x^2}{2} - 2 \frac{x^3}{3} \right) \\
 &= \frac{1}{4} (3x^2 - x^3)
 \end{aligned}$$

$$\begin{aligned}
 x \geq 2, \quad F(x) &= \int_{-\infty}^0 f(x) dx + \int_0^2 f(x) dx + \int_2^{\infty} f(x) dx \\
 &= \int_0^2 f(x) dx = \int_0^2 \frac{3}{8} (4x - 2x^2) dx \\
 &= \frac{3}{8} \times \frac{8}{3} = 1
 \end{aligned}$$

4) The diameter of an electric cable is assumed to be a r.v with pdf

$$f(x) = \begin{cases} kx^3(1-x), & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

a) Evaluate k

b) Obtain distribution function

c) Find $P(X < 2/3)$

Soln:-

$$\text{a) } \int_{-\infty}^{\infty} f(x) dx = 1 \Rightarrow \int_0^1 kx^3(1-x) dx = 1$$

$$\Rightarrow k \left. \frac{x^4}{4} \right|_0^1 - k \left. \frac{x^5}{5} \right|_0^1 = 1$$

$$\Rightarrow \frac{k}{4} - \frac{k}{5} = 1$$

$$\Rightarrow k = 20$$

$$b) -\infty < x \leq 0, f(x) = 0 \Rightarrow F(x) = 0$$

$$0 < x < 1, F(x) = \int_0^x 20x^3(1-x) dx$$

$$= 20 \left[\frac{x^4}{4} - \frac{x^5}{5} \right]$$

$$= 5x^4 - 4x^5$$

$$x \geq 1, F(x) = \int_0^1 20x^3(1-x) dx$$

$$= (5x^4 - 4x^5) \Big|_0^1 = 1$$

$$c) P(x < 2/3) = \int_0^{2/3} 20x^3(1-x) dx$$

$$= (5x^4 - 4x^5) \Big|_0^{2/3}$$

$$= 5 \left(\frac{16}{81} \right) - 4 \left(\frac{32}{243} \right) = \frac{112}{243}$$

$$= 0.4609$$

5) A conv X has the pdf $f(x) = 2x$,
 $0 < x < 1$. Find

(i) $P(X < 1/2)$ (ii) $P(X > 3/4 \mid X > 1/2)$

(iii) $P(X < 3/4 \mid X > 1/2)$

$$\begin{aligned} \text{(i)} \quad P(X < 1/2) &= \int_0^{1/2} f(x) dx = \int_0^{1/2} 2x dx \\ &= x^2 \Big|_0^{1/2} = \frac{1}{4} \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad P(X > 3/4 \mid X > 1/2) &= \frac{P(\{X > 3/4\} \cap \{X > 1/2\})}{P(X > 1/2)} \\ &= \frac{P(X > 3/4)}{P(X > 1/2)} \end{aligned}$$

$$P(X > 3/4) = \int_{3/4}^1 2x dx = x^2 \Big|_{3/4}^1 = 1 - 9/16 = 7/16$$

$$P(X > 1/2) = \int_{1/2}^1 2x dx = x^2 \Big|_{1/2}^1 = 1 - 1/4 = 3/4$$

$$P(X > 3/4 \mid X > 1/2) = \frac{7/16}{3/4} = \frac{7}{16} \times \frac{4}{3} = \frac{7}{12}$$

$$\begin{aligned}
 \text{(iii)} \quad P(X < 3/4 \mid X > 1/2) &= \frac{P(\{X < 3/4\} \cap \{X > 1/2\})}{P(X > 1/2)} \\
 &= \frac{P(1/2 < X < 3/4)}{P(X > 1/2)}
 \end{aligned}$$

$$\begin{aligned}
 P(1/2 < X < 3/4) &= \int_{1/2}^{3/4} 2x \, dx = x^2 \Big|_{1/2}^{3/4} = \frac{9}{16} - \frac{1}{4} \\
 &= \frac{5}{16}
 \end{aligned}$$

$$P(X > 1/2) = \int_{1/2}^1 2x \, dx = x^2 \Big|_{1/2}^1 = 1 - 1/4 = 3/4$$

$$P(X < 3/4 \mid X > 1/2) = \frac{5}{16} \times \frac{4}{3} = \frac{5}{12}$$